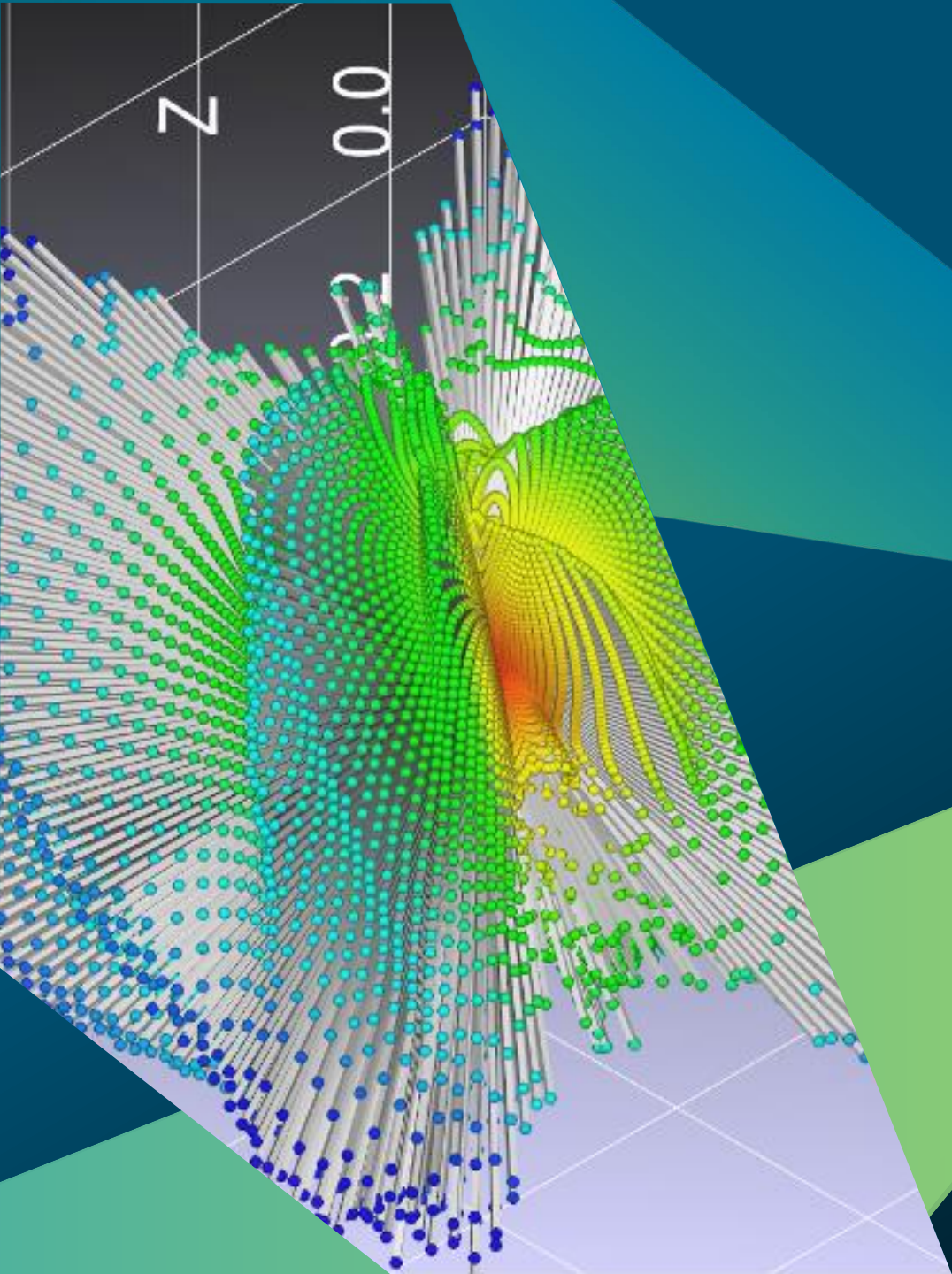




HEXAGON



CAEfatigue

Release 2025.1

What's new

(We recommend watching in slide show mode)

May 2025

New capabilities and enhancements in CAEfatigue 2025.1

Product Summary

CAEfatigue is the **fastest**, **easiest to use** and most **fully featured** fatigue tool available.

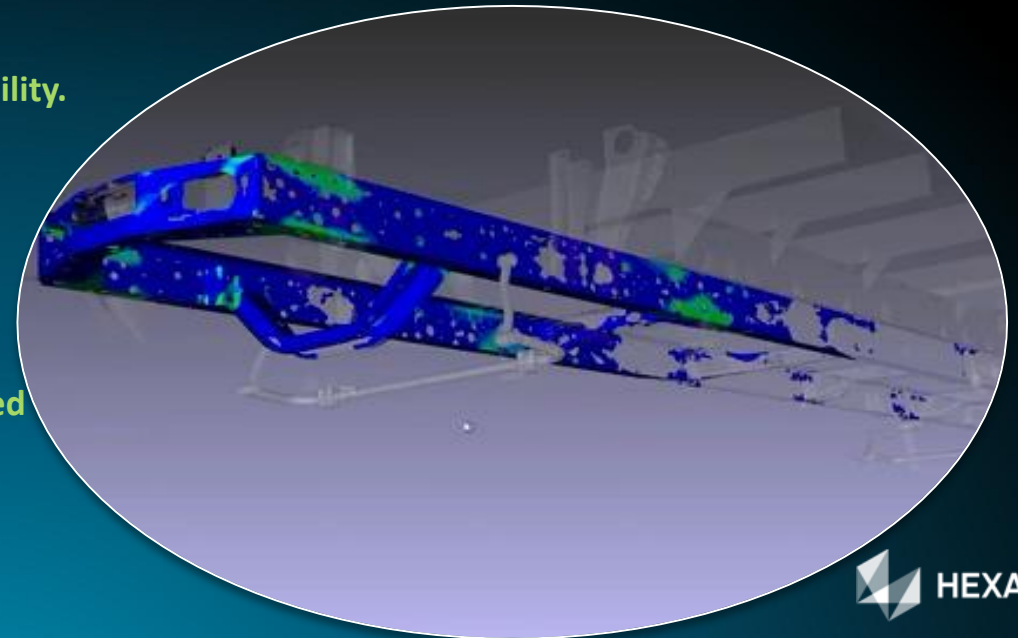
CAEfatigue is a random response and vibration fatigue solver for the frequency domain plus a fatigue solver for the time domain. It works with mixed random and deterministic loadings or standard time domain-based loadings to provide fatigue life and damage predictions, as well as several forms of response statistics. It is exceptionally fast, easy-to-use and capable of handling very large models. The software is sold in 4 different package configurations – Premium, Frequency, Random, and Time. As well as these fatigue tools the software includes state-of-the-art methods for loads conditioning (Fourier transforms), test design and correlation.

A unique FREQUENCY DOMAIN random response and vibration fatigue capability.

A state-of-the-art TIME DOMAIN fatigue solver.

A LOADS MANAGEMENT and TEST DESIGN toolkit including unique FFT loads conversion and loads manipulation tools.

A tool for checking how ROBUST a model is when model parameters are varied according to User specified distributions.



Major new solver features in CAEfatigue 2025.1

- Improved loads conditioning

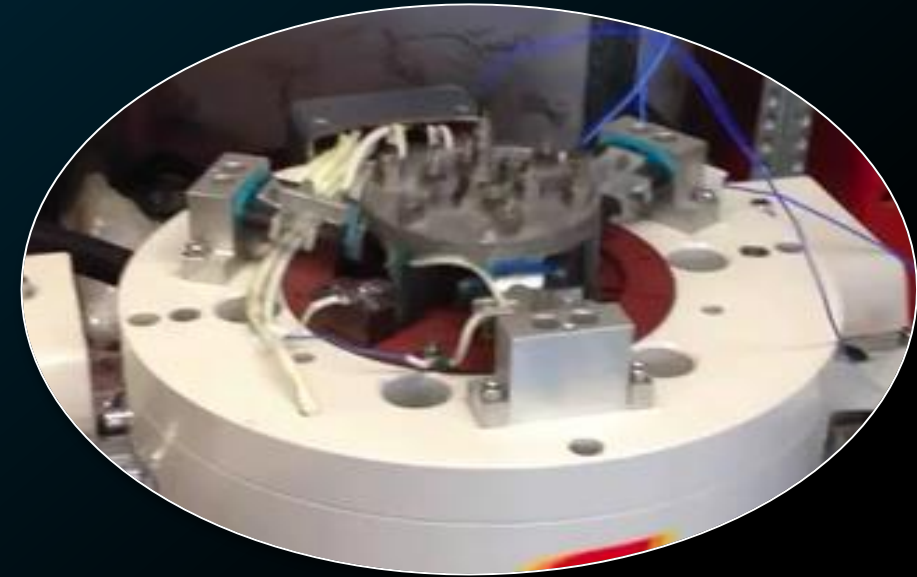
A new Method 8 is introduced for calculating the FFT window length T

- Improved surrogate load analysis using Adjustment Factors

This release introduces adjustment factors to Time2PSD, allowing for refined PSD calculations based on comparisons between time history and PSDM-derived damage estimations, improving accuracy and correlation.

- Seamweld Thickness Correction implementation

This release introduces a thickness correction for seamwelds, initially implemented for NEF, improving accuracy and addressing a known limitation in calculations.

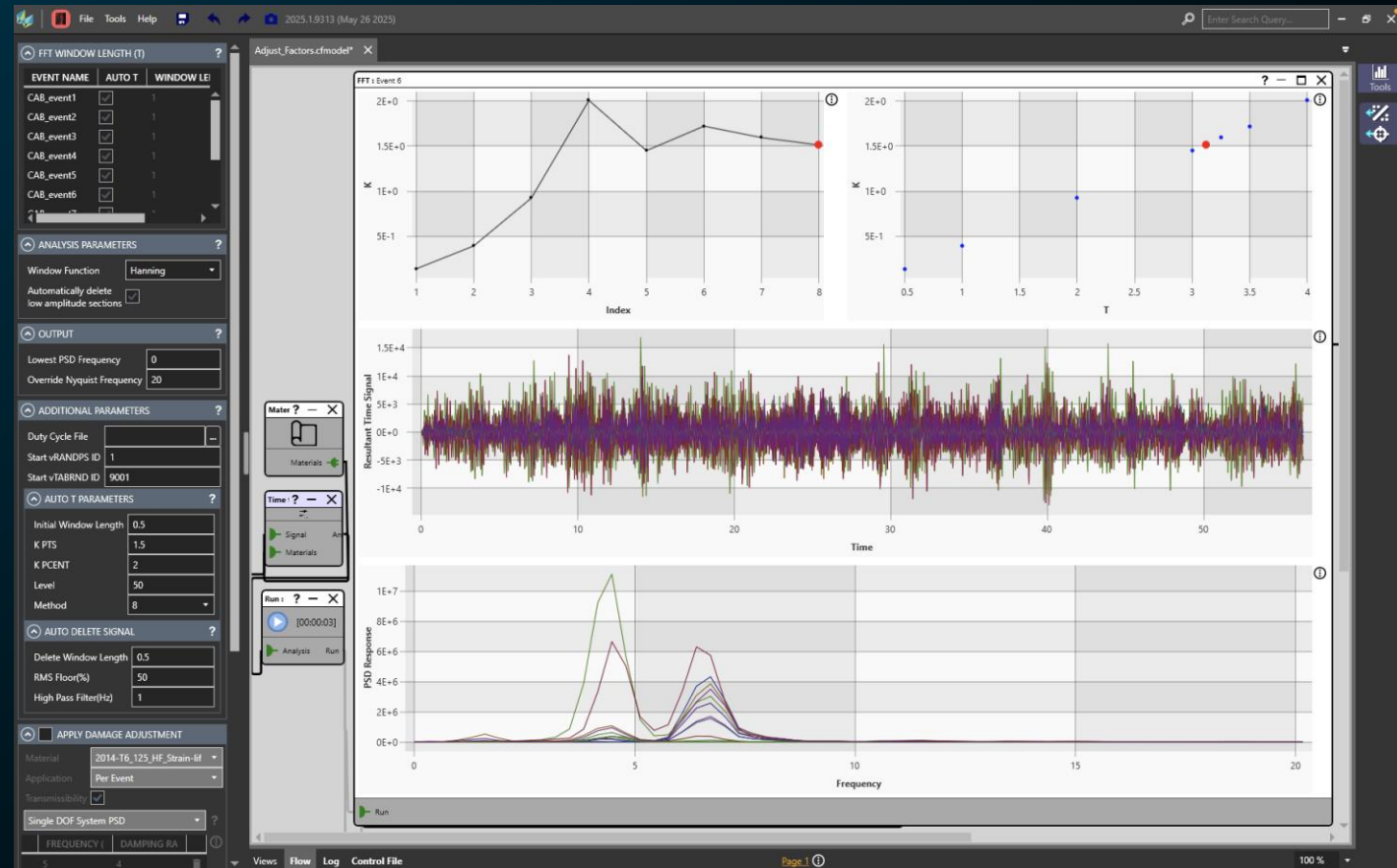


Improved Loads Conditioning (using a new Method 8 for calculating T)

TIME2PSD is used to convert time histories to PSDs using a Fourier transformation. The tool uses Welch's method to create an average PSD along a time signal of any length. Welch's method takes any number of PSD estimates and creates a linear average for the result. The key parameters are window shape, window length (T), and the amount of overlap (which fixes the number of windows used in the average).

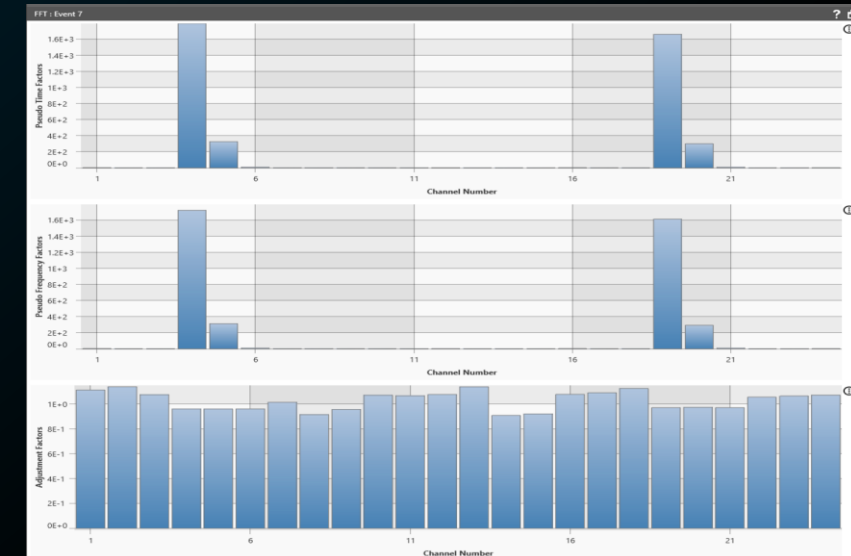
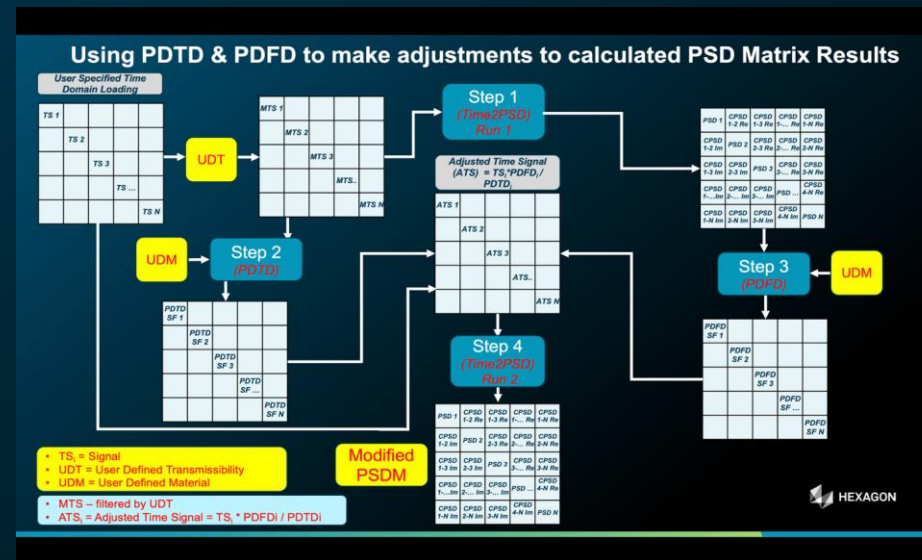
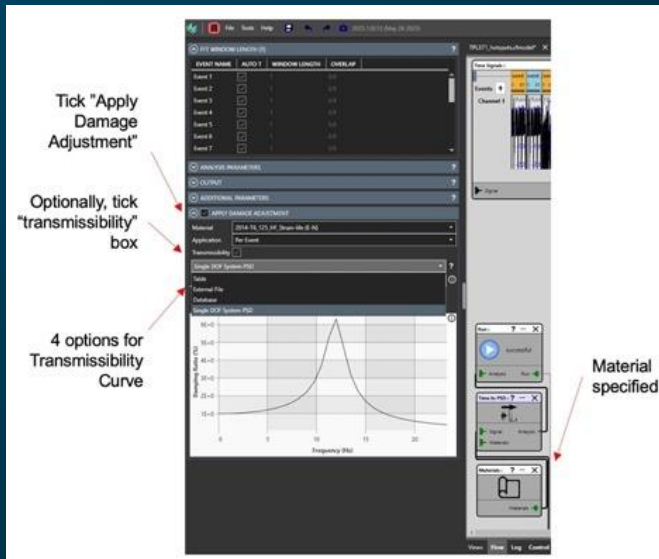
Included within this tool is a semi-automatic procedures (called autoT) to calculate the window length T. autoT has been extended (by adding Method 8) to make it insensitive to sample rate. As well as the direct PSD's, all of the cross PSD's (both real and imaginary) are calculated and stored in a so-called PSD Matrix.

Using the correct window length is a challenge. If too long a window is used the resultant PSD estimates will have high resolution but could contain too much scatter (because too few windows are used in the average). Using too short a window length will reduce scatter in the estimate but could result in poor resolution. The goal is to obtain a goldilocks result where both are just right. Method 8 with a default of 1.5 is a good starting estimate.



Improved surrogate load analysis using Adjustment Factors

The calculation of PSD's and PSD Matrices is vulnerable to errors because of divergence from the underlying assumptions (random, Gaussian, stationary). Because of this a new tool has been added to TIME2PSD to mitigate (correct for) these potential errors. The procedure is based on the use of pseudo damage estimates from the time and frequency domain results. The following procedure is followed.



Seamweld Thickness Correction implementation

Description:

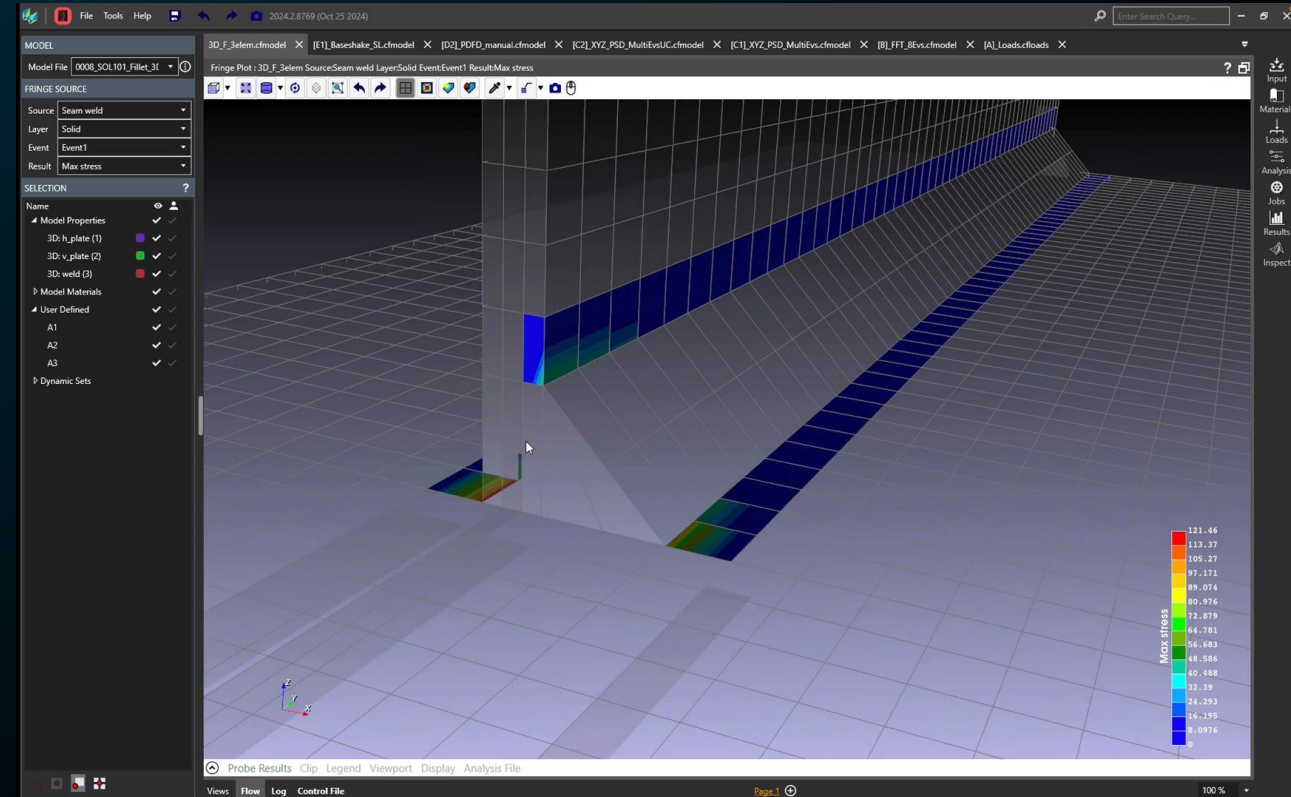
- This release introduces a thickness correction for seamwelds, initially implemented for NEF, improving accuracy and addressing a known limitation in calculations.

Benefits:

- Faster and more accurate weld fatigue results

Who needs it:

- Automotive, and GV designers and analysts wanting more efficient weld analysis



Other solver features in CAEfatigue 2025.1

Equivalent PSD Loading from Time Domain Strains: This release enables the calculation of equivalent PSD loading from time domain strains using transfer functions, allowing users to derive representative input PSDs from stress time histories measured at multiple locations.

Performance Improvement: Faster Frequency Domain Calculations: This release significantly improves the performance of frequency domain calculations, demonstrated by a reduction in runtime for typical pump assembly models from 13 minutes to 1 minute, achieved through parallelization and multithreading of the absmxpr calculation.

Material-Dependent Static Correction Implementation: This release implements material-dependent static correction, allowing users to define the static correction method within each material property, providing greater accuracy and control over fatigue analysis, as requested by Hendrickson and OshKosh.

Python and C++ API Enhancements: This release introduces a Python API built upon the existing C++ API and expands the scope of the C++ API, providing increased flexibility and programmability.

Support for Ansys .RDSP Files Added for Modal Analysis: This release adds support for importing modal participation factors from Ansys .RDSP files, providing an alternative format alongside .RSP and expanding compatibility with Ansys-based simulations.

Improved Solver Library Flexibility: This release enhances solver library flexibility, allowing users to select from multiple options including MKL, ACML, and others, with initial implementation focusing on externalizing the MKL library.

Test Problem and Verification Updates: This release includes updates to test problems and verification procedures, ensuring the accuracy and reliability of the software through rigorous testing and validation.

GUI specific & architecture enhancements in CAEfatigue 2025.1

Load Scheduler Usability Enhancements: This release improves the usability of the load scheduler by removing repetitions in event spreadsheets and consolidating the process of reading multiple response files for load time histories.

DSP Calculator Channel Manipulation Enhancements

This release enhances the DSP Calculator with the ability to combine multiple channels into a single channel, providing new functionalities for time history data manipulation, currently supporting channel addition.: **GUI Enhancements:**

Surface Resolve and Element ID Display: This release improves the GUI by displaying both 2D and 3D element IDs during surface resolve operations, aiding users in identifying the original 3D element associated with stress time history data.:

GUI Enhancement: Custom Material MMPDS Plotting: This release enhances the GUI to support MMPDS plotting for custom materials, enabling users to visualize material properties based on defined coefficients.

Loading Scenarios: Multiple Solver File Support: This release expands the functionality of Loading Scenarios to support multiple solver result files, allowing for more comprehensive analysis and data integration, currently applicable to non-OP2 file formats

GUI Enhancement: Template Insertion Functionality: This release introduces the ability to insert pre-defined templates into pages, allowing users to quickly add standardized control sets and leveraging existing templates from the File->New menu.

GUI Enhancement: Stress Amplitude Output and Custom Results: This release enhances the GUI by providing stress amplitude output for the H3D writer and enabling the creation of custom results using DSP calculator formulas within the fringe plotter, addressing a long-standing request from Scania.

GUI Enhancement: Automatic ZEROPSD Generation: This release enhances the GUI to automatically generate ZEROPSD/ZSDMODAL loads when an event contains only deterministic loads, simplifying the workflow and improving usability, with added UI tests to ensure functionality.

Fringe Plotter Enhancement: Improved Visibility Selection: This release enhances the Fringe Plotter with a visibility picker for rectangular selections, allowing users to select only visible elements, providing more precise control and improved usability for creating user groups.

Brown Miller Critical Plane Support Added to Process Flow: This release adds support for the Brown Miller Critical Plane stress combination method within the Process Flow, enabling users to define and utilize this approach for fatigue analysis.

Fatigue Tools 64-bit: The Fatigue Tools have been upgraded to 64-bit. This will make it easier to run on newer hardware.

Defect fixes in CAEfatigue 2025.1

ZSDMODAL Support Added: This release introduces support for the ZSDMODAL flag, enabling functionality for Single-Input MODALs with ZEROPSD contexts.

Frequency Domain Seam Weld Calculation Fix: This release corrects inaccurate results in frequency domain seam weld calculations when using op2 and psdmodal settings.

Default UTS Value Implemented: This release implements a default value of 400 MPa for UTS when left blank in the control file, aligning with existing documentation.

Time2PSD Subevent Handling Improvement: This release addresses a solver crash that occurred when Time2PSD subevents with a time range entirely outside the available time history data were processed, improving robustness and data handling.

Engineering Stress Output Enhancement: This release ensures that the 'TIMESTRN NO' parameter in the control file correctly outputs engineering stress even when EN materials are present, resolving a previous limitation.

Load Scheduler Export Fix: This release corrects an issue where the load scheduler would not write '0' values to the exported file when processing empty loads.

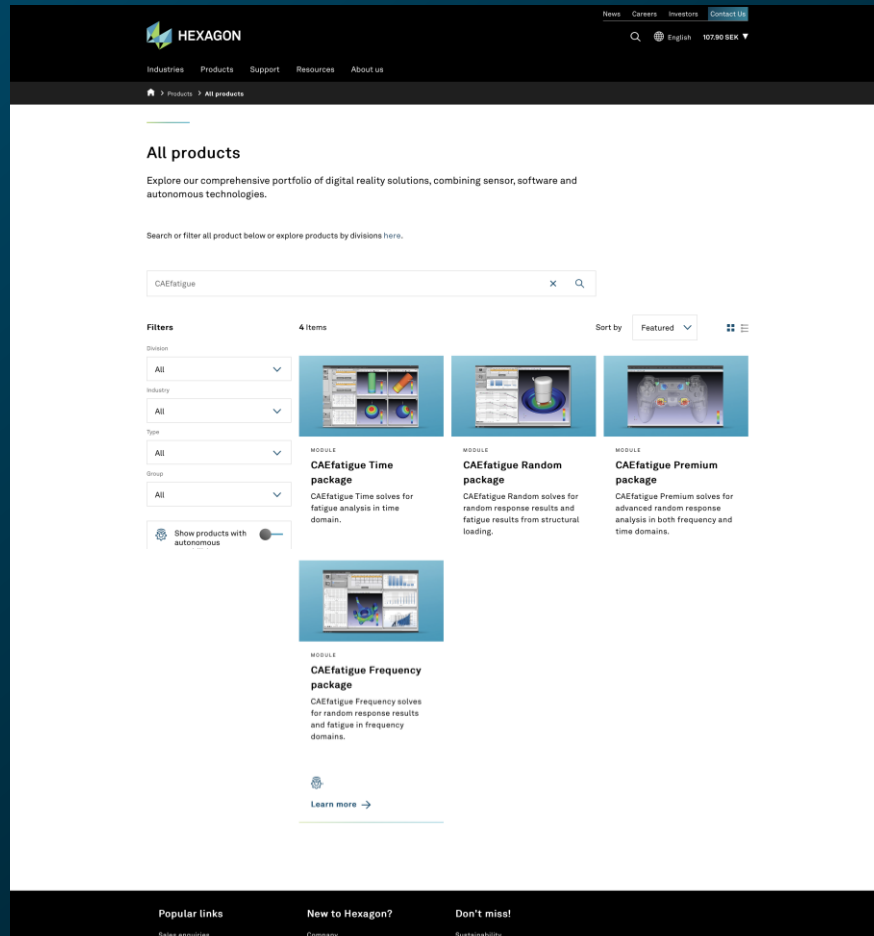
Load Time History Data Duplication Fix: This release resolves an issue where adding data by event to load timehistories from CAEfatigue results was causing field duplication, and includes checkboxes to maintain values.

Load Scheduler Time History Display Fix: This release resolves an issue where directly opening a timehistory file in the load scheduler failed to display the corresponding time histories, now addressed by redrawing all loads upon file loading.

Additional resources

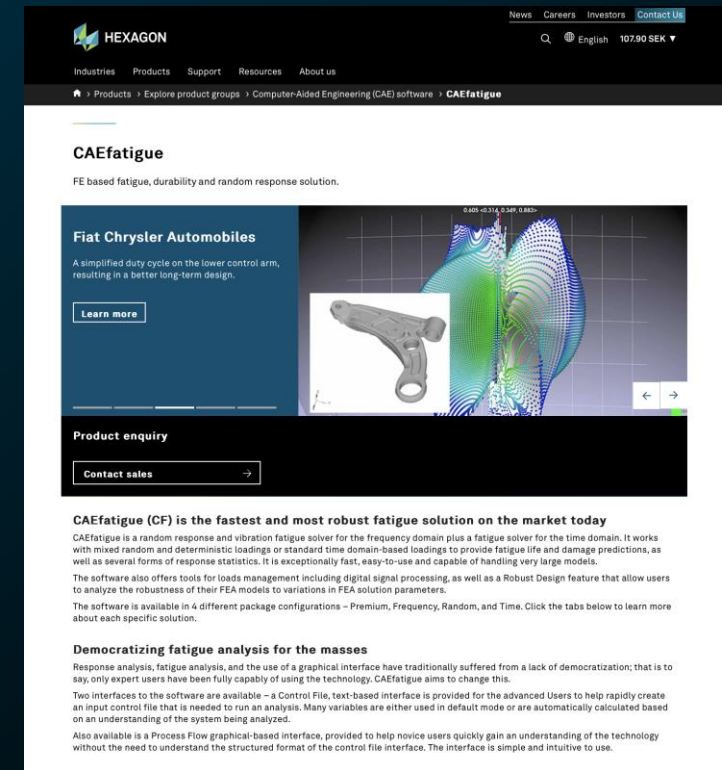
Support website

<https://simcompanion.hexagon.com/customers/s/article/CAEfatigue-Support-Home-Page>



CAEfatigue Homepage

<https://hexagon.com/products/product-groups/computer-aided-engineering-software/caefatigue>



Support Email:
support@mscsoftware.com



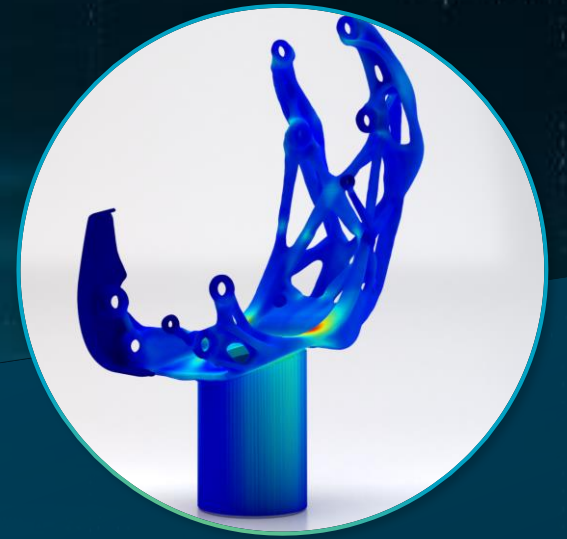
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Thank you

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